

The empirical recurrence for OEIS A317523: recovering a conjecture that is not written down

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Abstract

OEIS A317523 records “Empirical recurrence of order 59”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 138$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 59, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 59 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A317523 is “Number of $n \times 6$ 0..1 arrays with every element unequal to 0, 1, 2, 3, 4, 5, 6 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

32, 2048, 115446, 6475106, 364882328, 20564214798, 1158965686460, 65318143510572, ...

The entry states:

Empirical recurrence of order 59 (see link above)

As of the “Last modified” line on the live entry (Jul 30 2018 11:39:05, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4096$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 138$ of the 4096, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 138$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 276$ exact terms of the model returns order 59 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{59} c_i a(n-i),$$

c_1	56	c_{16}	−126718916385	c_{31}	−127941979376907	c_{46}	−5839699651840
c_2	75	c_{17}	199386566853	c_{32}	703816804625243	c_{47}	34591118803372
c_3	−2566	c_{18}	248375874934	c_{33}	283459879222372	c_{48}	−12234362823968
c_4	−29211	c_{19}	993212026206	c_{34}	−283736353075620	c_{49}	−12454506101456
c_5	−26405	c_{20}	3107305799698	c_{35}	−2511411143830	c_{50}	3163977074688
c_6	1040128	c_{21}	−3411690650763	c_{36}	−426740760474877	c_{51}	−5378265830592
c_7	3731144	c_{22}	−8360005725463	c_{37}	−378189535106430	c_{52}	−1080513540736
c_8	−12632016	c_{23}	−5676343374604	c_{38}	530875406355758	c_{53}	1781935184384
c_9	−121803449	c_{24}	−32991503421763	c_{39}	289954071482977	c_{54}	952187333632
c_{10}	−105043423	c_{25}	17460491198175	c_{40}	−101188892594410	c_{55}	508371533824
c_{11}	1399162478	c_{26}	145399156054638	c_{41}	74336230136532	c_{56}	−74802921472
c_{12}	3216711436	c_{27}	31557254766863	c_{42}	−164385859186682	c_{57}	−45664141312
c_{13}	869564277	c_{28}	−83148813224260	c_{43}	−182444987434958	c_{58}	−32114737152
c_{14}	−7311390839	c_{29}	−59796408522315	c_{44}	104893479780846	c_{59}	−19084083200
c_{15}	−80445981615	c_{30}	−372647862393495	c_{45}	44130217591248		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 59, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $59 + 4$ terms removed returns order 59 as well, so $\deg q_{\min} = 59 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $59 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 59 that this sequence satisfies — and that family turns out to have exactly one member.

References

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